WORKED SOLUTION

STEP III 2019, Question 1

Pure • Coupled Differential Equations

STEP III 2019, Question 1

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

For $k = 1$ the right-hand sides are exactly what you get by multiplying $x + \mathrm{i}y$ by a complex constant, so the pair of equations collapses into a single equation $\dot{z} = cz$ for $z = x + \mathrm{i}y$. That turns the motion into a rotation combined with exponential decay: a spiral. For $k = 0$ the coupling goes one way only: x doesn't depend on y, so we solve for x first and then feed it into the equation for y.

Solution

Part (i)

With $k = 1$ we have $\dot{x} = -x - y$ and $\dot{y} = x - y$. Put $z = x + \mathrm{i}y$. Then

$$\begin{aligned}\dot{z} = \dot{x} + \mathrm{i}\dot{y} &= (-x - y) + \mathrm{i}(x - y) \\ &= (-1 + \mathrm{i})(x + \mathrm{i}y) = (-1 + \mathrm{i})z,\end{aligned}$$

as we can confirm by expanding $(-1 + \mathrm{i})(x + \mathrm{i}y) = -x - \mathrm{i}y + \mathrm{i}x - y$. With $z(0) = 1$, the solution is

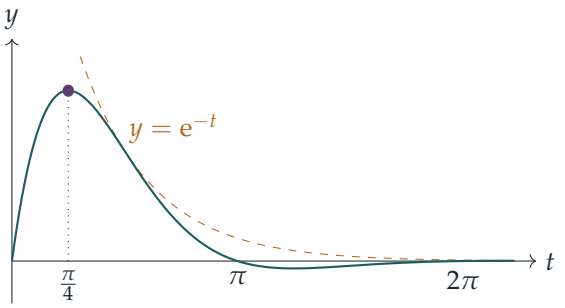
$$z = \mathrm{e}^{(-1+\mathrm{i})t} = \mathrm{e}^{-t}(\cos t + \mathrm{i}\sin t),$$

so

$$x = \mathrm{e}^{-t}\cos t, \qquad y = \mathrm{e}^{-t}\sin t.$$

(If you prefer to avoid complex numbers: differentiate the first equation and eliminate y to get $\ddot{x} + 2\dot{x} + 2x = 0$, with auxiliary roots $-1 \pm \mathrm{i}$, and use $x(0) = 1$, $\dot{x}(0) = -1$.)

Graph of y against t. This is a sine wave inside the envelope $\pm\mathrm{e}^{-t}$, so it's a damped oscillation. It is zero at $t = n\pi$, and its turning points are where $\dot{y} = x - y = \mathrm{e}^{-t}(\cos t - \sin t) = 0$, i.e. $t = \frac{\pi}{4} + n\pi$. Each successive peak is smaller by a factor $\mathrm{e}^{-\pi} \approx 0.04$, so after the first hump the oscillations are barely visible on a true scale.



The path. In polar form, $z = e^{-t}e^{it}$ has modulus $r = e^{-t}$ and argument $\theta = t$. So the path is the spiral

$$r = e^{-\theta}, \qquad \theta \geq 0,$$

starting at $(1, 0)$ and winding anticlockwise into the origin, getting there only as $t \to \infty$. At the start $\dot{x} = -1$ and $\dot{y} = 1$, so it leaves $(1, 0)$ moving up and to the left. Each turn shrinks the distance from the origin by a factor $e^{-2\pi} \approx 0.002$, so on a true scale only the first turn is visible.

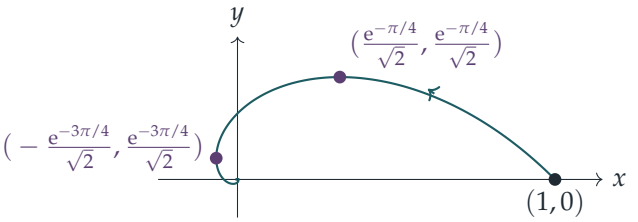
Greatest y. The turning points of y are at $t = \frac{\pi}{4} + n\pi$. At $t = \frac{\pi}{4}$ we get $y = e^{-\pi/4}/\sqrt{2} > 0$. For every $t \geq \frac{\pi}{2}$ we have $y \leq e^{-t} \leq e^{-\pi/2}$, and $e^{-\pi/2} < e^{-\pi/4}/\sqrt{2}$ because this is equivalent to $\sqrt{2} < e^{\pi/4} \approx 2.19$. On $[0, \frac{\pi}{2}]$ the only turning point is $t = \frac{\pi}{4}$. So the greatest value of y is at

$$\left(\frac{e^{-\pi/4}}{\sqrt{2}}, \frac{e^{-\pi/4}}{\sqrt{2}}\right) \approx (0.322,\ 0.322).$$

Least x. Now $\dot{x} = -x - y = -e^{-t}(\cos t + \sin t)$, which is zero when $\tan t = -1$, i.e. $t = \frac{3\pi}{4} + n\pi$. Where $x < 0$ we need $\cos t < 0$, which picks out $t = \frac{3\pi}{4} + 2n\pi$, giving $x = -e^{-t}/\sqrt{2}$; since e^{-t} is decreasing, this is most negative for the smallest such t, namely $t = \frac{3\pi}{4}$. (The least value must occur at one of these turning points, since $x(0) = 1 > 0$ and $x \to 0$ as $t \to \infty$.) So the least value of x is at

$$\left(-\frac{e^{-3\pi/4}}{\sqrt{2}}, \frac{e^{-3\pi/4}}{\sqrt{2}}\right) \approx (-0.067,\ 0.067).$$

Both points lie on the lines $y = \pm x$, as you'd expect: on the spiral, y is greatest where the path is heading in the x-direction, and x is least where it is heading in the y-direction.



Part (ii)

With $k = 0$ the first equation is $\dot{x} = -x$ on its own, so with $x(0) = 1$,

$$x = \mathrm{e}^{-t}.$$

Then $\dot{y} + y = \mathrm{e}^{-t}$. Multiplying by the integrating factor e^{t} gives $\frac{\mathrm{d}}{\mathrm{d}t}(y\mathrm{e}^{t}) = 1$, so $y\mathrm{e}^{t} = t + c$, and $y(0) = 0$ gives $c = 0$:

$$y = t\mathrm{e}^{-t}.$$

Ask yourself How do we turn this into an equation for the path?

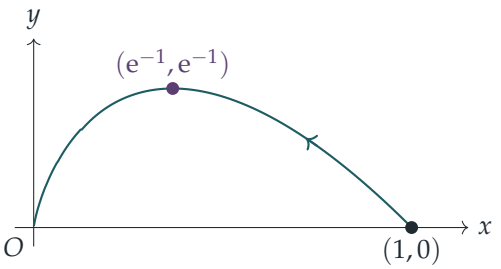
Eliminate t: since $x = \mathrm{e}^{-t}$, we have $t = -\ln x$, and so

$$y = -x \ln x, \qquad 0 < x \leq 1.$$

As t runs from 0 to ∞, x decreases from 1 towards 0, so the particle starts at $(1, 0)$ and moves leftwards along this curve, approaching the origin as $t \to \infty$.

Features for the sketch:

- $\frac{\mathrm{d}y}{\mathrm{d}x} = -\ln x - 1$, which is zero at $x = \mathrm{e}^{-1}$. So the highest point is $(\mathrm{e}^{-1}, \mathrm{e}^{-1})$, reached at $t = 1$.
- At $(1, 0)$ the gradient is -1, matching $\dot{x} = -1$, $\dot{y} = 1$ at $t = 0$.
- As $x \to 0^{+}$, $x \ln x \to 0$, so $y \to 0$; but the gradient $-\ln x - 1 \to +\infty$, so the curve comes into the origin tangent to the y-axis.
- $y > 0$ for $0 < x < 1$, so the whole path lies above the x-axis.



Notice the contrast with part (i): with $k = 0$ there's no longer any rotation, so the particle never crosses an axis. It simply curves round into the origin.

Checks

- Substitute back: for $x = \mathrm{e}^{-t}\cos t$, $\dot{x} = -\mathrm{e}^{-t}\cos t - \mathrm{e}^{-t}\sin t = -x - y$, as required; $\dot{y}$ works the same way. Both solutions also satisfy $x(0) = 1$, $y(0) = 0$.

- Stepping the equations forward numerically in small time steps reproduces both solutions, and confirms the greatest y in part (i) is about 0.3224 at $t \approx 0.785$, the least x is about -0.0670 at $t \approx 2.356$, and in part (ii) y peaks at $1/\mathrm{e} \approx 0.3679$ at $t = 1$.
- Sanity check on (ii): $y = t\mathrm{e}^{-t}$ is the familiar "repeated root" form, which makes sense since the matrix of the system has the repeated eigenvalue -1.

Common slips

- Taking the least value of x at $t = -\frac{\pi}{4}$ or $\frac{7\pi}{4}$: $\dot{x} = 0$ there too, but $x > 0$ there (it's a local maximum). Check the sign of $\cos t$.
- Forgetting that the spiral only reaches the origin as $t \to \infty$, or drawing it winding clockwise: $\theta = t$ increases, so it winds anticlockwise.
- In (ii), missing the vertical tangent at the origin: $y \to 0$ but $\mathrm{d}y/\mathrm{d}x \to \infty$.

Beyond STEP

Any linear system $\dot{\mathbf{x}} = A\mathbf{x}$ is governed by the eigenvalues of A. Here $A = \begin{pmatrix} -1 & -k \\ 1 & -1 \end{pmatrix}$ has eigenvalues $-1 \pm \sqrt{-k}$: for $k = 1$ they are the complex pair $-1 \pm \mathrm{i}$, giving a *stable spiral*; for $k = 0$ there is the repeated eigenvalue -1, giving a *degenerate node*; and for $k < 0$ you'd get two real eigenvalues. Classifying fixed points this way (phase-plane analysis) is a core first-year topic. Boyce and DiPrima, *Elementary Differential Equations*, covers it with plenty of pictures.

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